Statistical and Computational Models

Lecture 13: Approximating the posterior

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Roadmap

Priors and posteriors

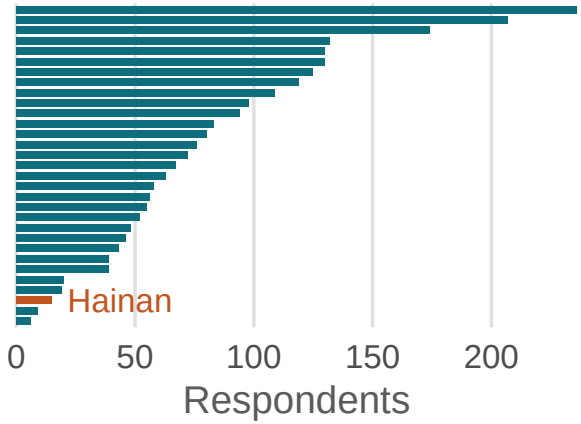
Approximating the posterior

Bayesian regression

Trust in local government

Suppose we survey 2,500 adults across China's 31 provinces:

province	age	educ	hukou	trust
Yunnan	35-59	Secondary	Rural	1
Shanxi	18-34	College	Rural	0
Guangdong	35-59	Secondary	Urban	1
Jiangsu	18-34	College	Urban	1
Guangdong	35-59	College	Urban	0
Hainan	18-34	College	Urban	1



Y_i = 1 if respondent *i* trusts local government, 0 otherwise.

Target: π, the share of **all** Hainan adults who trust local government.

Without seeing data: what's your best guess for π, and how sure are you?

Estimating trust in Hainan

But we do have some data! Suppose for Hainan we have 15 respondents, 12 of whom said "yes". What natural estimators might we try?

HAINAN ONLY

Subset to Hainan, and find the MLE, the sample share:

$$\hat{\pi} = r/n = 12/15 = 0.80$$

$$\text{se}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/15} = 0.103$$

$$\hat{\pi} \pm 1.96\,\text{se} = [0.598,\ 1.002]$$

Assumes a **random sample** of Hainan's adults. Our interval is wide because we only use limited data.

ALL PROVINCES

Pool all 2,500 respondents, and find the MLE:

$$\hat{\pi} = 1{,}449/2{,}500 = 0.58$$

$$\text{se}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/2{,}500} = 0.010$$

$$\hat{\pi} \pm 1.96\,\text{se} = [0.560,\ 0.599]$$

Assumes Hainan is **like the others**. Precise, but Hainan is 0.6% of the data, so it may miss Hainan.

Bias and variance

Re-run the survey 1,000 times, as random samples of the same size, and average. We know π, since we simulated it.

Estimator	π	One draw	Average	Spread (sd)
No pooling				
Hainan (n = 15)	0.67	0.80	0.68	0.12
Guangdong (n = 236)	0.56	0.50	0.56	0.03
Complete pooling				
Hainan (n = 2,500)	0.67	0.65	0.65	0.01

No pooling, Hainan vs. Guangdong: both are right on average, but with 15 respondents any one survey can land far off. High **variance**: alone, a small province **overfits**.

Complete pooling: almost no spread, but always off, by Hainan's gap from the national share. High **bias**: pooling **underfits**.

Bayesian inference

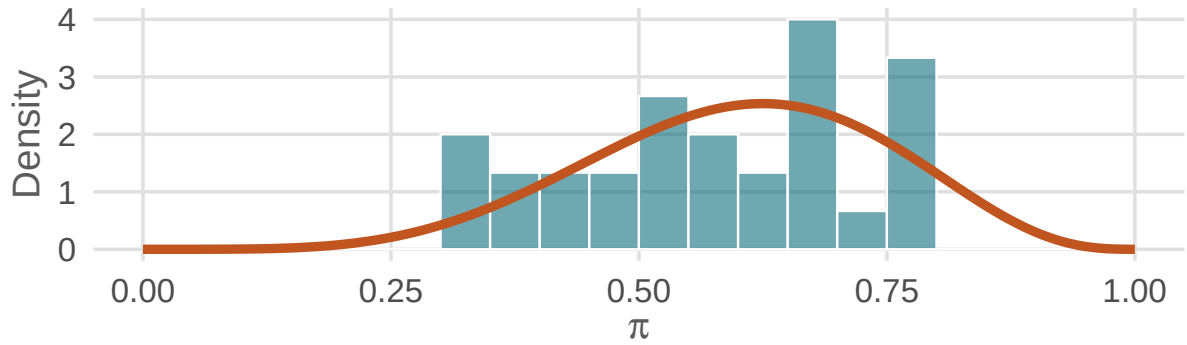
Maximum likelihood assumed a probability model for Y only. Suppose we **also** assume a distribution for the parameter, before we see the data.

1. **Assume** a model for the data (here, as before, $Y_i \sim \text{Bernoulli}(\pi)$), and a **prior** $p(\pi)$: how plausible each value of π is.
2. **Guess** values of the parameter, and weight each by its prior, times how likely the data would be had it been the true value.
3. **Learn** a distribution for the parameter given the data, the **posterior** $p(\pi \mid \mathbf{Y})$, not a single number.

That is, π is still fixed and unknown. The prior and the posterior describe **our uncertainty** about it.

A prior for Hainan

Your guesses, and how sure you were of them, are a prior. Another source is the other 30 provinces' sample shares (bars): center 0.58, sd 0.15.



To turn the bars into a prior, we need a distribution on $[0, 1]$. A typical choice is the **Beta** distribution, whose parameters a and b act like ones and zeros already seen:

$$\pi \sim \text{Beta}(a, b), \qquad \mathbb{E}[\pi] = \frac{a}{a + b}$$

Choose a and b to match the bars: Beta(6, 4) has center 0.60, sd 0.15 (curve). So it is **worth 10 respondents**: as if we had already asked 10 Hainan adults, and 6 said yes. Against Hainan's 15, that is a lot; against Guangdong's 236, very little.

Bayes' rule

How do we combine the prior with the data? **Bayes' rule**:

$$\underbrace{p(\pi \mid \mathbf{Y})}_{\text{posterior}} = \frac{\overbrace{p(\mathbf{Y} \mid \pi)}^{\text{likelihood, } L(\pi;\mathbf{Y})} \overbrace{p(\pi)}^{\text{prior}}}{\underbrace{p(\mathbf{Y})}_{\text{normalizing constant}}}$$

$$p(\pi \mid \mathbf{Y}) \propto L(\pi;\mathbf{Y})\, p(\pi)$$

Why can we drop $p(\mathbf{Y})$?

It does not depend on π: it only makes the posterior integrate to one.

Guessing π, with a prior

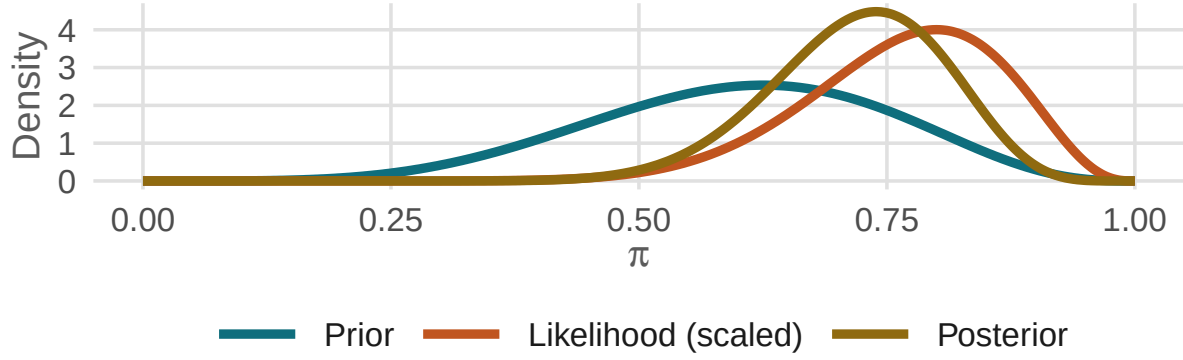
Suppose π could only be one of five values. Our Beta(6, 4) prior at each, rescaled to sum to one:

π	Prior	$L(\pi; \mathbf{Y})$	Prior $\times$ L	Posterior
0.5	0.23	3.1e-05	7.2e-06	0.03
0.6	0.30	1.4e-04	4.2e-05	0.17
0.7	0.27	3.7e-04	1.0e-04	0.41
0.8	0.16	5.5e-04	8.7e-05	0.35
0.9	0.04	2.8e-04	1.0e-05	0.04
Sum	1.00		2.5e-04	1.00

The posterior peaks at 0.7, not 0.8 (the MLE): 15 respondents are not enough to overrule a prior worth 10.

Unlike the MLE, we learn how **plausible** each value is, not just which is best.

The Beta-Binomial



Now let π take **any** value in $[0, 1]$, not just five. Hainan's prior, Beta(6, 4), becomes a curve.

Add the likelihood, rescaled to area one. Where will the posterior sit?

Between the two, nearer the likelihood: 15 respondents against a prior worth 10.

With a Bernoulli likelihood, a Beta prior gives a Beta posterior: the prior is **conjugate**.

$$\pi \mid \mathbf{Y} \sim \text{Beta}(a + r,\ b + n - r) = \text{Beta}(18, 7)$$

The prior and the sample size

Posterior means under three priors, for Hainan and for Guangdong:

Prior	Worth	Prior sd	Hainan (n = 15)	Guangdong (n = 236)
None (MLE)			0.800	0.466
Beta(1, 1)	2	0.29	0.765	0.466
Beta(6, 4)	10	0.15	0.720	0.472
Beta(12, 8)	20	0.11	0.686	0.477

The priors move Hainan up to 11 points from the MLE, but Guangdong only 1. Why?

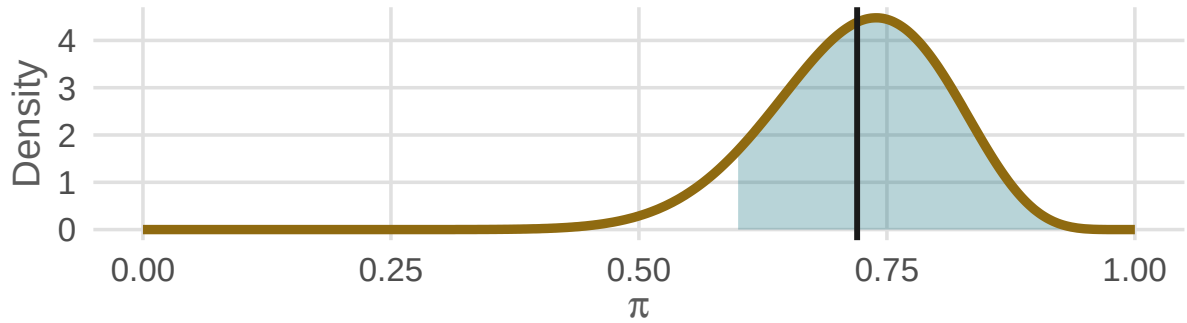
$$\mathbb{E}[\pi \mid \mathbf{Y}] = \underbrace{\frac{a+b}{a+b+n}}_{\text{weight on prior}} \cdot \frac{a}{a+b} + \underbrace{\frac{n}{a+b+n}}_{\text{weight on data}} \cdot \frac{r}{n}$$

Read: a prior worth more is **spikier**, it carries more information.

But the prior is an **assumption**, and it can be wrong!

Summarizing the posterior

The posterior is a distribution, which we can **summarize**:



Point estimate: the posterior mean, 0.72 (solid line).

Credible interval: the middle 95% of the posterior, [0.53, 0.87].

Read: given our model and prior, there is a 95% probability that π lies in it. Different to a confidence interval, whose 95% describes the procedure, over **repeated samples**.

Probabilities: the area above 0.6 (shaded), $\mathbb{P}(\pi > 0.6 \mid \mathbf{Y}) = 0.90$, the chance that over 60% of Hainan's adults trust.

Roadmap

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Approximating the posterior

Bayesian regression

The normalizing constant

Dropping $p(\mathbf{Y})$ tells us what $p(\pi \mid \mathbf{Y})$ is proportional to, but not the posterior itself. For that we need

$$p(\mathbf{Y}) = \int_0^1 L(\pi; \mathbf{Y})\, p(\pi)\, d\pi$$

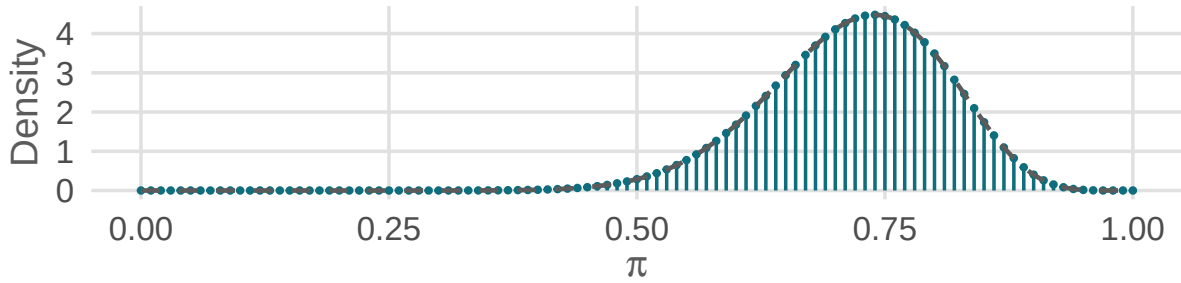
For a proportion with a Beta prior, conjugacy lets us skip this integral: the posterior is a Beta, a known distribution.

For many models, such as the logit, no standard prior is conjugate and the integral for $p(\mathbf{Y})$ cannot be solved, so the posterior has no closed form.

Then we are left having to **approximate** the posterior.

Grid approximation

We have already tried an approximation: a rough **grid** of five guesses.



We can do better by making many finer guesses, dividing prior times likelihood by its sum.

But grids grow fast: 100 points for one coefficient, 100×100 for two, etc.

Much of the grid falls where the posterior is near zero – can we spend our effort more efficiently?

Coefficients	Evaluations	Approx. time
1	100	under a second
2	10,000	under a second
4	10^8	1.1 hours
7	10^{14}	127 years

Moving through the posterior

Rather than evaluate a fixed grid, we can walk through Hainan's posterior one step at a time. At step t, we are at $\pi^{(t)}$. Where next?

An optimizer only moves uphill, to the peak. To learn a **distribution**, we move uphill whenever we can, but also **downhill** sometimes.

1. Start somewhere: $\pi^{(0)}$
2. Propose a random step: $\pi^* = \pi^{(t)} + \varepsilon, \quad \varepsilon \sim N(0, s^2)$
3. Compare prior times likelihood: ratio $= \dfrac{L(\pi^*; \mathbf{Y})\, p(\pi^*)}{L(\pi^{(t)}; \mathbf{Y})\, p(\pi^{(t)})}$
4. $\pi^{(t+1)} = \pi^*$ with probability $\min(1, \text{ratio})$; otherwise $\pi^{(t+1)} = \pi^{(t)}$. Repeat.

Prior times likelihood cannot give the posterior at π, but it can compare two values of π: in the ratio, $p(\mathbf{Y})$ **cancels**. This is the **Metropolis** algorithm.

A Metropolis chain for Hainan

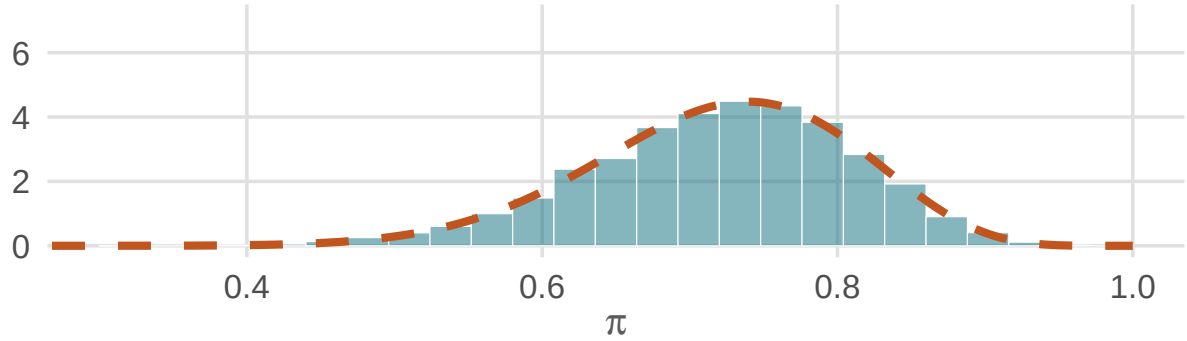
Back to Hainan: prior Beta(6, 4), steps from $N(0, 0.1^2)$, a start at 0.5. The first 1,000 draws are usually uninformative about the posterior, so we drop them ('burning in'). The next six, moving if $u <$ ratio, with u uniform on $[0, 1]$:

t	$\pi^{(t)}$	π^*	ratio	u	Move?	$\pi^{(t+1)}$
1,001	0.586	0.560	0.667	0.485	yes	0.560
1,002	0.560	0.480	0.199	0.302	no	0.560
1,003	0.560	0.431	0.055	0.605	no	0.560
1,004	0.560	0.424	0.044	0.158	no	0.560
1,005	0.560	0.671	3.798	0.812	yes	0.671
1,006	0.671	0.701	1.186	0.205	yes	0.701

1,006: uphill (ratio above 1), so we always move.

A Metropolis chain for Hainan

Keep going, and collect the draws:



All 19,000 draws (after burning the first 1,000).

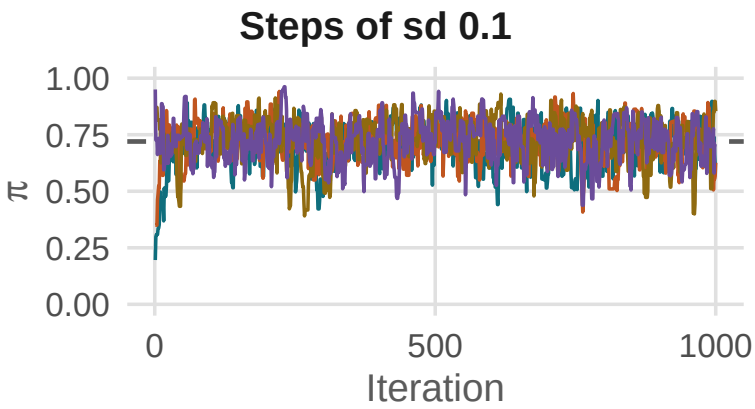
Run long enough, the draws settle on the posterior.

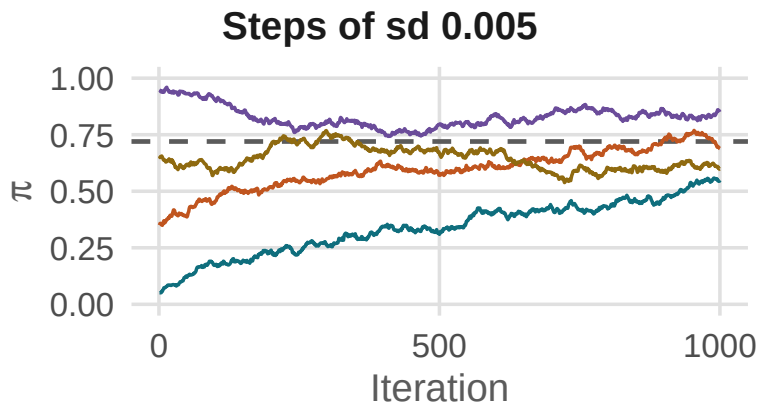
Note that a rejected step records the current value **again**, so the chain spends more time where the posterior is higher.

Where the chain goes next depends only on where it is now: a **Markov chain**. We compute means and intervals from the draws, not from a formula: **Monte Carlo**. Together, **Markov chain Monte Carlo** (MCMC).

Checking convergence

Typically, we run four chains from different starting values. A **trace plot** shows each chain's draws by iteration.





The chains meet fast, then mix like a fuzzy caterpillar. They have **converged**.

With tiny steps, they have **not converged** after 1,000 iterations.

R-hat compares the spread of all the draws pooled with the spread within each chain. If the chains explore the same posterior, the two match and R-hat is near 1 (1.00 left, 3.86 right).

A common rule: below 1.05 suggests convergence; then **combine** the kept draws.

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Modeling trust in Hainan

Consider a logit using the Hainan data only: `trust ~ hukou + age`.

	Estimate	Std. error
`(Intercept)`	2.1	1.4
`hukouUrban`	-0.5	1.4
`age35-59`	-0.9	1.4
`age60+`	16.0	3,956.2

The data suffer from **separation**: Only one respondent in Hainan is aged 60+. That respondent answered 'yes', so the larger $\beta_{\texttt{60+}}$, the closer their predicted Pr(trust) gets to 1, and the likelihood keeps rising: the MLE of $\beta_{\texttt{60+}}$ is $+\infty$.

`glm()` stops at 16, once the gains are tiny, with an se of 3,956. Its Pr(trust) for an urban 60+ adult is 1.00, from one respondent.

A coefficient of 16 on the logit scale is implausible. We could encode this belief in a prior, one that puts little weight on huge values and so keeps $\beta_{\texttt{60+}}$ finite.

Modeling trust in Hainan

For regression, the priors go on the coefficients: each coefficient β_j gets its own prior, with center m_j and spread s_j:

$$\beta_j \sim N(m_j, s_j^2)$$

Because a coefficient can be any real number, we need a prior that lives on the real number line. We choose a normal prior, rather than a Beta (which lives on $[0, 1]$).

We can pick any prior we want. Here we compare two: a weak prior centered on zero, and one centered on the other provinces' `glm()` estimates, with sd 0.5.

	Weak	From the other provinces
`(Intercept)`	$N(0, 2.5^2)$	$N(0.57, 0.5^2)$
`hukouUrban`	$N(0, 2.5^2)$	$N(-0.69, 0.5^2)$
`age35-59`	$N(0, 2.5^2)$	$N(0.32, 0.5^2)$
`age60+`	$N(0, 2.5^2)$	$N(0.55, 0.5^2)$

Modeling trust in Hainan

The posterior has no closed form, so we use Metropolis, with β in place of π: four chains, 1,000 burned from each, R-hat $\leq$ 1.01.

Prior	β_{60+} (truth 0.55)	Pr(trust), urban, 60+
None (`glm()`)	16.0 (se 3,956)	1.00
Weak, $N(0, 2.5^2)$	0.95 [-3.00, 5.43]	0.84 [0.24, 1.00]
From the other provinces	0.64 [-0.32, 1.65]	0.75 [0.45, 0.93]

The weak prior downweights extreme values, so β_{60+} is finite. But its interval is nearly as wide as the prior's, because one respondent provides little information.

The prior from the other provinces provides much more information, so the posterior is much narrower, and close to the truth. As with Beta(6, 4), with little data the prior does the work, so its information must come from **somewhere**.

Fitting Bayesian models in R

How to fit Bayesian models in R:

Package	A call for our logit	You supply
{rstanarm}	`stan_glm(trust ~ hukou + age, data = hainan, family = binomial, prior = normal(0, 2.5, autoscale = FALSE))`	formula; priors from a menu
{brms}	`brm(trust ~ hukou + age, data = hainan, family = bernoulli, prior = set_prior("normal(0, 2.5)"))`	formula; any prior
{cmdstanr}, {rstan}	a model written in Stan's own language	everything

All of these run **Stan**, a language for writing Bayesian models: we state the data, parameters, priors and likelihood, and Stan builds the sampler. It uses Hamiltonian Monte Carlo (which we won't cover), which follows the slope of the log posterior to take long steps. Most are still accepted, so each draw is worth far more. Its burn-in, 'warm-up', also tunes the sampler.

Choosing a prior

A posterior probability is only as good as its prior. Four ways to limit or justify the prior's influence:

- **Let the data swamp it**: with enough data, the prior barely matters. Our three priors moved Guangdong by at most 1 point, Hainan by up to 11.
- **Check sensitivity**: report the answer under several priors. If a conclusion survives every reasonable one, the prior is not driving it.
- **Use weak priors**: $N(0, 2.5^2)$ on a logit coefficient rules out only absurd values, odds ratios beyond $e^5 \approx 148$.
- **Estimate it**: a tight prior is justified when we already know a lot about this parameter. Data from other provinces count only as far as provinces are alike, and a hierarchical model learns how alike from the data.

Frequentist intervals rest on assumptions too: Hainan's confidence interval assumes a random sample. A prior makes one more assumption explicit.

References

- Efron, B. & Hastie, T. (2016) *Computer Age Statistical Inference* – **ch. 3**.
- Johnson, A. A., Ott, M. Q. & Dogucu, M. (2022) *Bayes Rules! An Introduction to Applied Bayesian Modeling* – **ch. 6**; chs. 2–5 and 7 as background. bayesrulesbook.com.